

Longitudinal Data Analysis Involving Binary Outcomes

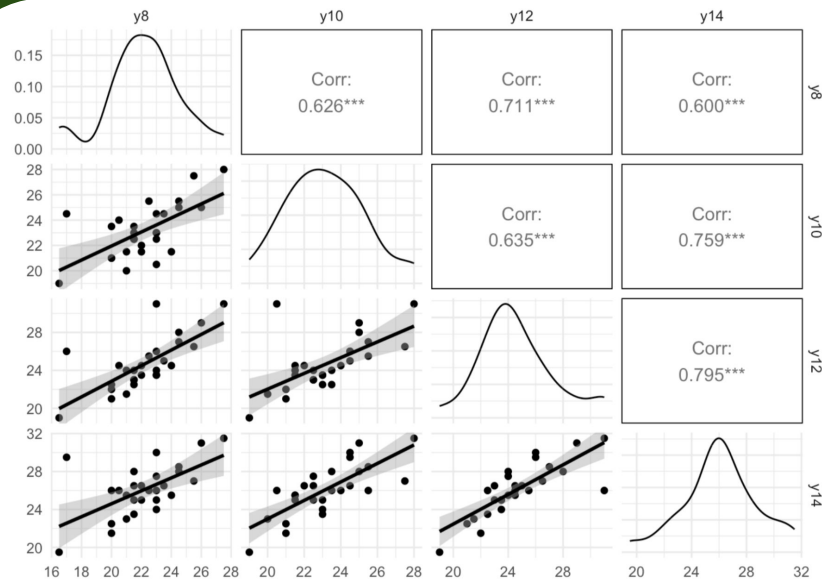
Introduction to Biostatistics & Fundamentals of Biostatistics and Epidemiology

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What is Longitudinal Data Analysis?

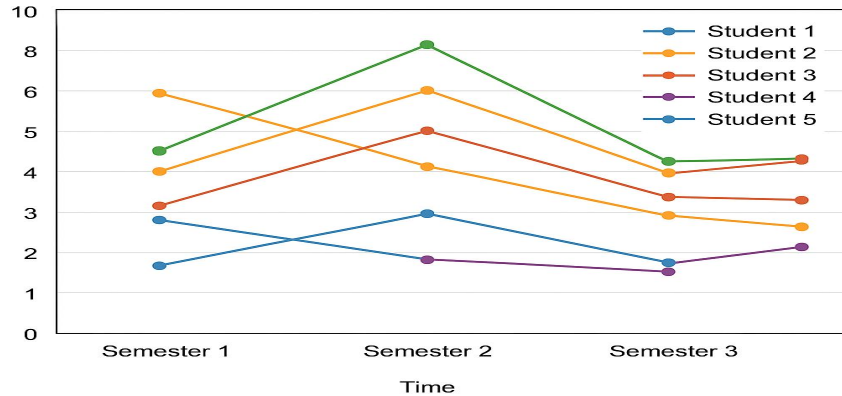
Longitudinal data refers to *repeated measurements* of the same variables—e.g., Health outcomes, biomarkers—collected from the same subjects over multiple time points. Unlike cross-sectional data, which is more similar to a single snapshot in time, longitudinal data *tracks within-subject changes*.

Longitudinal analysis examines data collected over multiple time points to study temporal changes, widely applied in medicine, economics, and social sciences to track trends like disease progression or economic shifts. Using methods such as mixed-effects models, growth curve modeling, and time series analysis, it evaluates individual and group-level dynamics, distinguishes within- and between-subject variability, and explores causal relationships.

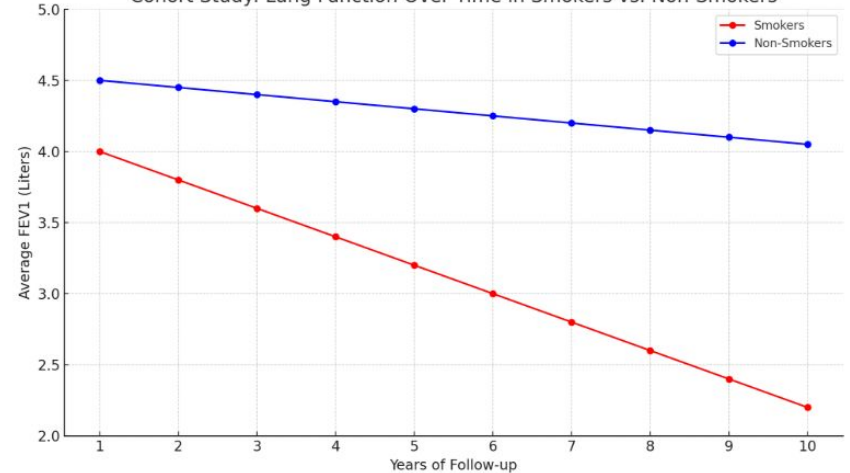
While advantageous for capturing evolving processes, challenges include managing missing data, addressing time-dependent structures, and isolating individual trajectories. It is particularly valuable in medical research for tracking treatment responses, disease trajectories, and patient-specific variability, offering insights static data cannot provide.

There are 2 Main Types of Longitudinal Studies

Tracking Stress Levels in University Students Over Time



Cohort Study: Lung Function Over Time in Smokers vs. Non-Smokers



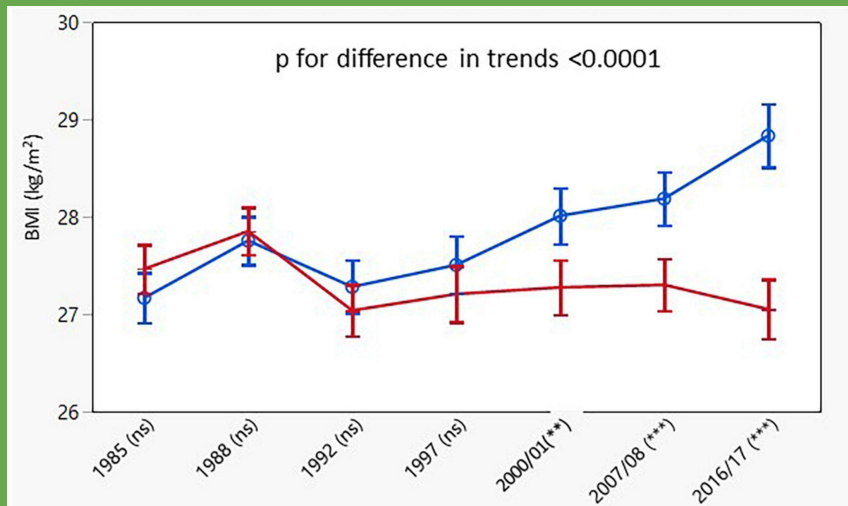
Panel Studies

- Tracks the same group of individuals (panel) over time to study changes or trends within them.
- The same individuals are followed throughout the study.
- Examines changes or trends within the same group of people.
- Allows detailed tracking of individual changes over time.
- Can suffer from participant dropout or bias due to repeated questioning.

Cohort Studies

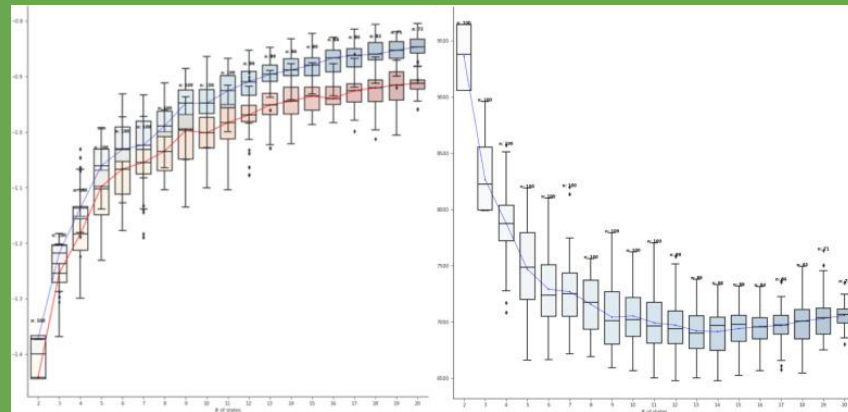
- Observes a group of individuals (cohort) sharing a common characteristic or exposure over time.
- Participants are selected based on a shared characteristic (e.g., birth year, exposure to a risk).
- Studies the relationship between exposures and outcomes (e.g., smoking and lung cancer).
- Ideal for studying long-term effects of exposures or life course events.
- May require large sample sizes and long follow-up periods for rare outcomes.

Why do we often encounter it in medical research?



Longitudinal trends in blood pressure, prevalence, awareness, treatment, and control of hypertension in the Czech population. [2]

LDA tracks BMI trends *over time* in the same population, adjusting for age, to identify meaningful shifts in health patterns. It tests if sex-specific trends differ significantly and accounts for repeated measurements, ensuring robust insights into *why* and *how* BMI changes occur.



Model performance (vertical axis) as a function of the number of latent states K (horizontal axis). [3]

LDA tracks how outcomes (e.g., symptoms, treatment effects) change over time in the same people. It shows if improvements are lasting or temporary, accounts for individual differences, and avoids misleading "snapshot" comparisons. It's ideal for seeing trends—whether a treatment works better after weeks vs. days, even if some patients miss follow-ups. This helps doctors understand real-world effectiveness.

What is Iterative Graphical Data Analysis?

IGDA is a systematic process of repeatedly visualizing data through graphical methods before, during, and after statistical modeling to uncover patterns, validate assumptions, and diagnose model performance.

Modeling →	Pre	During	Post
Purpose	Identify trends, outliers, and missing data.	Verify assumptions like linearity, homoscedasticity (constant variance), or correlation structures.	Diagnose model fit and residual behavior.
Example	Spaghetti plots (individual trajectories over time) reveal between–and <i>within</i> –subject variability (e.g., erratic blood pressure trends in a clinical trial).	Plotting <i>residuals</i> versus <i>fitted values</i> to check for nonlinearity or heteroscedasticity.	Residual vs. fitted plots or QQ-plots to assess <i>normality</i> and identify <i>systematic biases</i> .

Importance of Iterative Graphical Analysis

Visualization of
Data



Iterative
Improvement



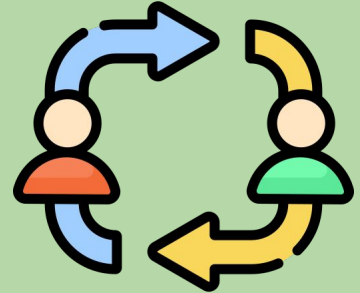
Debugging and
Optimization



Decision
Making



Feedback
Loops



Visualization of Data

- Allows for the visual representation of complex data sets.
- Helps in identifying patterns, trends, and outliers that might not be apparent from raw data.
- Easier to observe and interpret the underlying patterns, trends, and anomalies within data.
- Essential for a wide range of applications.



Iterative Improvement

- Supports the iterative process of model refinement and hypothesis testing.
- Each iteration can be visualized to compare with previous iterations or theoretical predictions, allowing for incremental improvements.
- Allows researchers and analysts to track model adjustments.



Debugging and Optimization

- Aids in pinpointing issues within a data set or model.
- Essential for optimizing parameters in engineering designs and algorithms.
- Ensuring that systems operate efficiently and accurately.



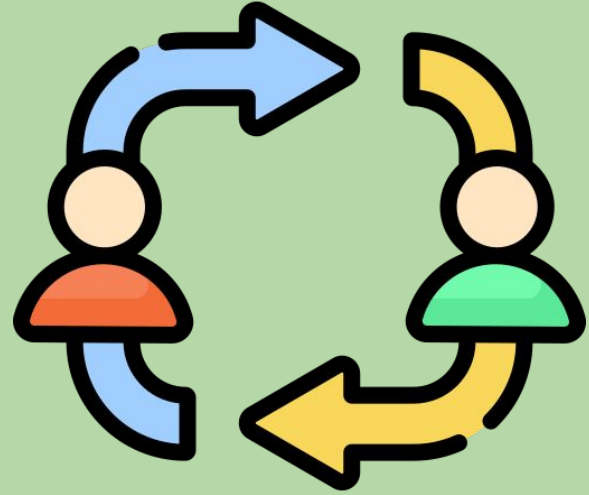
Decision Making

- Supports decision-making processes by providing graphical representations of scenarios and outcomes.
- Can be used to simulate different conditions and predict outcomes.
- Enables users/stakeholders to simulate different conditions.



Feedback Loops

- Enables the establishment of feedback loops in system design and analysis, crucial for dynamic systems and control engineering.
- Helps in the real-time adjustment of systems and processes based on visual feedback.
- Important for maintaining system stability and performance.



Methods of Longitudinal Data Analysis

Traditional Methods

Repeated Measures ANOVA (RMANOVA) is used to examine mean differences in a dependent variable measured over time on the same subjects.

Types:

1. One-way RMANOVA: One within-subjects factor (e.g., time).
2. Two-way RMANOVA: Includes an additional factor (e.g., time * treatment).

Example: Measuring blood pressure of 10 patients before, during, and after a treatment.

Assumptions

Normality: Outcome variable should be normally distributed.

Sphericity: Equal variance of differences between all time points.

Homogeneity of Variance: Equal variances across groups.

Independence: Each observation must be independent. Random sampling is essential.

Constant Time Points: Measurements must be taken at consistent intervals.

No missing values allowed!

STRENGTHS

RMANOVA

WEAKNESSES

Within-Subject Control: Accounts for individual differences by modeling within-subject correlations, increasing statistical power.

Flexible Factor Testing: Evaluates both within-subject (e.g., time) and between-subject (e.g., treatment groups) effects, including interactions (e.g., time $\times$ group).

Simplicity and Accessibility: Familiar framework for ANOVA users, with user-friendly implementations in common statistical software (SPSS, R, SAS).

Efficiency: Tests multiple time points simultaneously, avoiding separate pairwise comparisons.

Trend Analysis: Supports polynomial contrasts (e.g., linear, quadratic trends) for time-related patterns.

Sphericity Assumption: Requires equal variances of differences between time points. Violations necessitate corrections (e.g., Greenhouse-Geisser) or alternative methods.

Missing Data Handling: Uses listwise deletion for incomplete cases, risking bias and reduced power if data is not missing completely at random (MCAR).

Time as Categorical: Treats time as discrete, limiting flexibility for modeling continuous/time-varying trends or uneven intervals.

Balanced Design Requirement: Struggles with unbalanced data or varying measurement times, unlike mixed-effects models.

Normality Assumption: Sensitive to non-normal residuals or outliers; non-parametric alternatives are less developed.

Complex Interactions: Interpretation becomes challenging with higher-order interactions (e.g., multi-group $\times$ multi-time-point designs).

Scalability Issues: Performance declines with many time points due to increased comparisons and sphericity violations.

Modern Methods

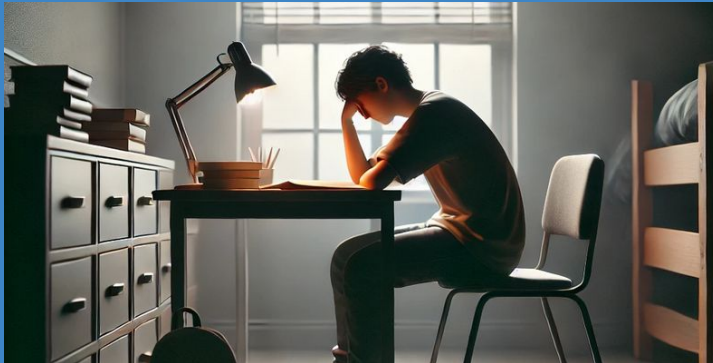
Aspect	GEE (Generalized Estimating Equation)	Mixed Effects Regression (MER)
Focus	Population-averaged effects	Subject-specific effects
Assumptions	Working correlation	Random effects distribution
Use Case	Marginal inference (e.g., public health)	Individual prediction (e.g., clinical trials)
Handling of missing data	Missing completely at Random (MCAR)	Missing at Random (MAR)
Computational complexity	1. Simple	1. Complex
	2. Computationally efficient	2. Computationally intensive
	3. Quasi-likelihood method	3. Maximum likelihood methods

PROS, LIMITATIONS	GEE	MER
PROS	1. GEE can account for other predictor variables in the model. 2. It can handle variation in time points at which the outcome is measured. 3. GEE assumed missing values as MCAR & provides an estimate	1. LMER can account for other predictor variables in the model. 2. It can handle variation in time points at which the outcome is measured. 3. LMER assumed missing values as MAR & provides an estimate. 4. Model diagnostics can be compared.
LIMITATIONS	1. Only Population average effects are estimated. 2. Model diagnostics cannot be compared.	Complex, computationally intensive

Feature	RMANOVA	LMER / Mixed Models	GEE
Handles missing data	✗	✓	✓
Allows unequal time points	✗	✓	✓
Individual-level variation	✗	✓ (random effects)	✗
Population-level estimation	✓	✓	✓ (focus)
Time-varying covariates	✗	✓	✓
Effect size estimation	✗	✓	✓

Implementation Using R

- Data set retrieved from [Kaggle](#)
 - A statistic analyzing depression levels found in students
 - Dataset involves →
 - Demographic information (age, gender)
 - Academic performance (grades, attendance)
 - Lifestyle habits (sleep patterns, exercise, social activities)
 - Mental health history
 - Responses to standardized depression scales.



Implementation Using R

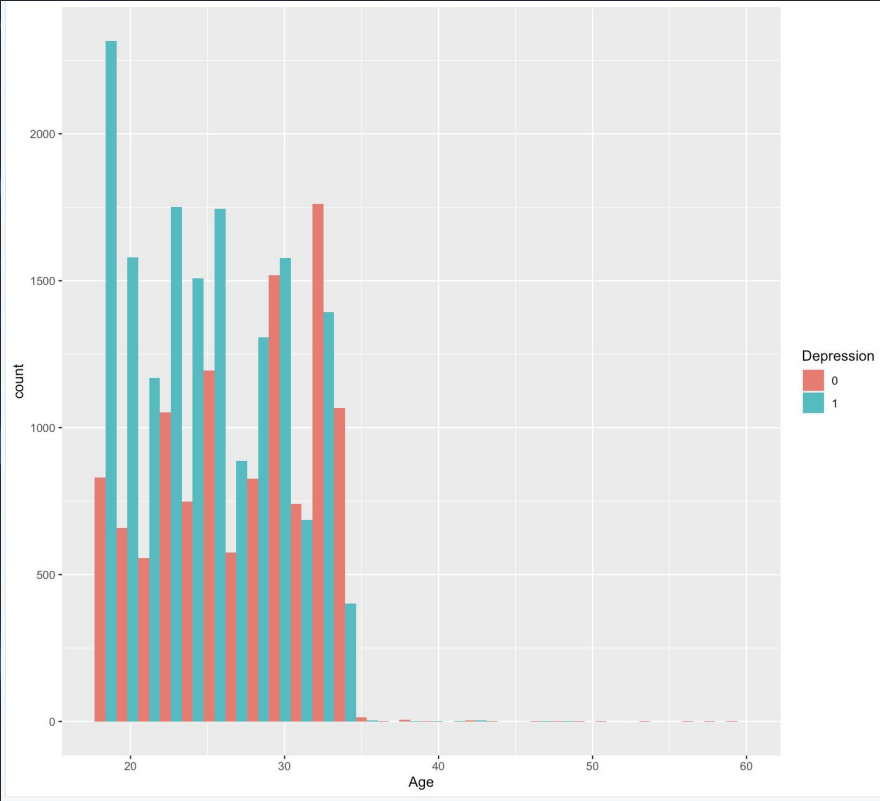
```
Call:
glm(formula = Depression ~ . - id - City - Profession, family = binomial(),
    data = data)

Coefficients:
(Intercept)          -3.577548    0.282743 -17.646 < 2e-16 ***
GenderMale           -0.011199    0.036944  -0.303  0.7618
Age                 -0.119314    0.004932 -24.193 < 2e-16 ***
'Academic Pressure'  0.842437    0.014925  56.445 < 2e-16 ***
'Work Pressure'      0.198144    0.600801   0.330  0.7416
GPA                 0.061268    0.012432   4.928 8.30e-07 ***
'Study Satisfaction' -0.240909    0.013606 -17.706 < 2e-16 ***
'Job Satisfaction'   0.090622    0.490757   0.185  0.8535
'Sleep Duration'7-8 hours 0.017527    0.051803   0.338  0.7351
'Sleep Duration'Less than 5 hours 0.364720    0.051268   7.115 1.12e-12 ***
'Sleep Duration'More than 8 hours -0.257612    0.054349  -4.740 2.14e-06 ***
'Sleep Duration'Others -0.228326    0.643808  -0.355  0.7229
'Dietary Habits'Moderate 0.471408    0.045219  10.425 < 2e-16 ***
'Dietary Habits'Others  0.923214    0.776921   1.188  0.2347
'Dietary Habits'Unhealthy 1.091055    0.046385  23.522 < 2e-16 ***
DegreeB.Com         -0.155373    0.141511  -1.361  0.1735
DegreeB.Ed          -0.059477    0.108856  -0.546  0.5848
DegreeB.Pharm       -0.116321    0.132631  -0.877  0.3805
DegreeB.Tech        -0.076369    0.121382  -0.629  0.5292
DegreeBA           -0.257435    0.142904  -1.801  0.0716
DegreeBBA          -0.031742    0.130746  -0.229  0.8190
DegreeBCA          -0.068428    0.112979  -0.606  0.5447
DegreeBE           0.008208    0.141175   0.058  0.9536
DegreeBIM          -0.149636    0.128325  -1.166  0.2436
DegreeBSC          0.123622    0.130360   0.948  0.3430
DegreeClass 12     -0.188751    0.094907  -1.989  0.0467 **
DegreeLLB          0.130680    0.140009   0.933  0.3506
DegreeLLM          0.265042    0.158271   1.675  0.0940 .
DegreeM.Com        -0.046274    0.138411  -0.334  0.7381
DegreeM.Ed         0.067887    0.134633   0.504  0.6141
DegreeM.Pharm      -0.005014    0.148936  -0.034  0.9731
DegreeM.Tech       -0.110188    0.125690  -0.877  0.3807
DegreeMA           -0.119881    0.148983  -0.805  0.4210
DegreeMBA          -0.195758    0.148012  -1.323  0.1860
DegreeMBS         0.138217    0.138664   0.997  0.3189
DegreeMCA          0.043389    0.123967   0.350  0.7263
DegreeMD           0.053485    0.149345   0.358  0.7202
DegreeME          -0.144439    0.230767  -0.626  0.5314
DegreeMIM         -0.026043    0.219928  -0.118  0.9057
DegreeMSc         -0.141543    0.118985  -1.190  0.2342
DegreeOthers       0.521191    0.481141   1.083  0.2787
DegreePhD          0.115580    0.153942   0.751  0.4528
'Have you ever had suicidal thoughts?'Yes 2.510349    0.039145  64.130 < 2e-16 ***
'Work/Study Hours'  0.117902    0.004799  23.681 < 2e-16 ***
'Financial Stress'  0.553728    0.013344  41.496 < 2e-16 ***
'Family History of Mental Illness'Yes 0.246844    0.036459  6.770 1.28e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 37855  on 27897  degrees of freedom
Residual deviance: 19449  on 27852  degrees of freedom
(3 observations deleted due to missingness)
AIC: 19541

Number of Fisher Scoring iterations: 6
```



Implementation includes:

1. Loading necessary libraries & dataset
2. Data preparation
3. Model fitting
4. Model Summary
5. Diagnostics and visualization

Implementation Using R

- **Longitudinal data analysis** → studying the same objects over multiple time points.
 - Applied to dataset,, the analysis helps to observe developments/changes in students' depression levels.
- **Binary outcomes** → refers to the two possible states of depression status (depressed or not depressed, coded as 1 or 0).

In the implementation context:

- **Multiple Observations Over Time** → change in data provides basis for longitudinal analysis
- **Binary Outcomes** → Logistic regression to model the probability of depression
- **Statistical Techniques** → Mixed-effects models to handle repeated measures.

Conclusion

- Longitudinal data analysis allows us to track changes over time, offering insights into individual trajectories and group-level trends.
- Key methods include mixed-effects models, GEE each suited for specific research questions.
- Proper handling of within-subject correlation, missing data, and time-varying covariates is essential for valid inference.
- These techniques are powerful tools in medical, social, and biological sciences, where time-based measurements are critical.
- Continued advancement in computational tools and model development is expanding the potential of longitudinal research.

Appendix: Citations

- [1] Cri, Ale. "Review Longitudinal." *RPubs*, RPubS, https://rpubs.com/alecri/review_longitudinal.
- [2] "Cardiovascular Medicine." *Frontiers in Cardiovascular Medicine*, vol. 9, 2022, <https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2022.1033606/full>.
- [3] "PMC." *PubMed Central (PMC)*, National Center for Biotechnology Information, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8075441/>.
- [4] "Longitudinal Data Analysis Methods - A Primer." *ResearchGate*, https://www.researchgate.net/publication/369740121_Longitudinal_data_analysis_methods_-_a_primer.
- [5] "Student Depression Dataset." *Kaggle*, <https://www.kaggle.com/datasets/hopesb/student-depression-dataset?resource=download>.